

EPS Technical Manual

An Overview of the EPS Main Board Digital Section

GENERAL

The main board of the EPS is based on a 68000 16-bit microprocessor along with the EPROMs, Static RAMs, Dynamic RAMs and I/O chips common to many microcomputers and the random logic needed to connect the various circuits.

MEMORY

OSROM

There are 64K bytes of permanent memory (which can be addressed by the 68000 as 32K words, where a word is 16 bits of data) using two, 27256-type 32K x 8 EPROMs.

OSRAM

There are 64K bytes (or 32K words) of high-speed RAM using two, 62256-type 32K x 8 Static RAM chips.

Sound RAM

A 256K x 13 RAM array is provided for sound storage using 41256-type 256K x 1 Dynamic RAMs. The 13-bit wide memory is needed as samples for the EPS are stored as 13-bit words. This Sound RAM is expandable in two ways. A 2X expander, which provides an additional 256K x 13 of memory, can be plugged into the memory expansion port. This supplies a total of 512K x 13 of RAM which can be used to store sounds and/or sequences. Alternately, a 4X expander, which provides an additional 1Meg x 13 of memory can be installed. With the 4X expander, the standard 256K x 13 of internal memory becomes available exclusively for the sequencer, while the RAM in the expander is used for sounds.

DUART

The EPS uses a 68681 DUART (*Dual Universal Asynchronous Receiver/Transmitter*) for most of its I/O functions. This chip is substantially the same as the 2681 DUART used in the ESQ-1 synthesizer, but has been adapted to the 68000 bus architecture. The DUART provides two asynchronous serial ports, 8 independent output pins, 6 independent input pins and a timer which is used as the master timing reference for the system by generating a periodic interrupt to the 68000 (much the same as the timer used in the ESQ-1). One of the serial ports is used exclusively for MIDI, while the other serial port communicates with the KPC microprocessor on the Poly-pressure keyboard. The output pins are used to control the disk drive, select the various analog inputs and to provide a source of data requests to the DMA controller for sampling (more later). The input pins are used to communicate with the disk drive and to receive external timing references which set the communication rates of the two serial ports.

Pin Assignments

There are six DUART inputs:

IP5 and **IP2** should both have a 1 MHz clock signal on them for the KPC port.
IP4 and **IP3** should both have a 500 KHz clock signal on them for the MIDI port.
IP1 is the Disk Change Status line, which should pulse high whenever a disk is removed from the drive.
IP0 is the Disk Status line, which should go high whenever a disk is in the drive and the drive is selected.

There are eight DUART outputs:
OP7 is the MIC/LINE level selector. When low, the sampling input is set for Line level, when high, it is set for Mic level.
OP6, **OP5** and **OP4** are the analog select lines, AN0, AN2 and AN1. They are used to select which analog input is to be digitized (wheels, sliders, etc).
OP3 is the SREQ (*Sample REQuest*) output. It is normally low, but during sampling, the DUART timer output appears on this line, which is wire-OR'd with the disk request line by transistor Q3. This sends a steady stream of DMA requests at the selected sample rate. If during sampling, there isn't a steady stream of DMA requests from this output, something is wrong.

OP2 is the MUTE line. When low the audio outputs are allowed to pass signal (such as when playing), when high the audio outputs are muted (such as when sampling).
OP1 is the Side Select control line for the floppy disk.
OP0 is the Disk Select control line for the floppy disk. It will go high whenever the disk is being accessed.

The MIDI port uses the standard optocoupler circuit for input and 7407 buffers for output. The KPC port goes directly to the keyboard connector, where it ultimately connects to the serial port of the 68HC11 on the KPC board.

FLOPPY DISK CONTROLLER

The EPS uses the same 1772 FDC (*Floppy Disk Controller*) as used in all of Ensoniq's disk-based products. The interface and buffers are identical, with the addition of a *Side Select* line (which is used to select which side of the double-sided drive is being accessed) and a *Disk Change* status line, which informs the microprocessor whenever a disk is removed or inserted into the drive. The IRQ and DRQ lines from the FDC are routed through the DMA controller, to allow for disk transfers under either microprocessor or DMA control. During any disk access, these lines will be moving. During a DMA disk access, DMA transfers will occur in response to these lines.

DMA CONTROLLER

The 68440 DMA controller (or DMAC - *Direct Memory Access Controller*) is the device which allows the EPS to load in new sounds while it is playing previously loaded sounds. All other sampling keyboards go dead while loading because the main microprocessor is used to take incoming information from the disk controller and place it in sound memory. The microprocessor is simply too busy handling the disk data to make any sound. The DMAC is a chip specifically designed to transfer data from one place to another at very high speed -- much faster than a microprocessor ever could. To accomplish this, when the disk controller signals the DMAC that a byte of information is available, the DMAC briefly halts the 68000, takes over the bus, gets the data from the disk controller and stores it in the appropriate location in memory, then releases the bus back to the 68000. This happens so fast that the 68000 is hardly disturbed and is able to continue doing whatever it was doing before the DMAC took over. Under normal circumstances, the time stolen from the 68000 has no effect on performance. The DMAC is also used when sampling to transfer the incoming samples from the DOC II chip to sound memory.

The 68440 was specifically designed to work with the 68000 and is therefore efficient and quick. The only support circuits needed are two 74LS373 eight-bit latches. The 68440 multiplexes it's address and data on the same pins and these latches allow the two busses to be seperated using control signals generated by the DMAC. When the DMAC is in control of the bus, it acts exactly like the 68000, generating all of the normal 68000 control signals. It can access any device that the 68000 can in the system memory map.

DMA Control Signals

The DMAC shares almost all of the 68000 control lines. For example, during a DMA operation, the address bus, data bus and control signals (R/W, Upper and Lower Data Strobes, etc.) all act exactly as if the 68000 were generating them, only they are actually coming from the DMAC. Therefore, any reference in this text to a "processor access" applies equally to a DMA access. In addition to the common control lines, there are special control lines which allow the DMAC and processor to hand off control to one another. These lines are BR (*Bus Request*), with which the DMAC requests control of the bus, BG (*Bus Grant*), with which the 68000 informs the DMAC that the bus is available and BGACK (*Bus Grant Acknowledge*), with which the DMAC informs the 68000 that it has taken over the bus. These lines should only ever move during a DMA transfer, that is, either a DMA disk access (such as loading or saving a sound or loading a disk overlay) or during sampling. If any of these lines are low or moving at other times, something is wrong. Additionally, the HIBYTE line should always be high.

DOC II

The DOC II is Ensoniq's second generation sound chip. It has many advances over the original DOC chip which provide superior fidelity and features.

General

The sample encoding scheme uses more bits of resolution -- in this case 13 bits, as compared with 8 bits in the DOC I. This greatly reduces distortion and improves the signal-to-noise ratio by 32 times (which is twice the accuracy and half the noise of a 12-bit sampler). Secondly the DOC II contains a high-speed digital signal processor which affects the sound in two ways.

To understand the following processes, it helps to understand that the DOC chips use a fixed playback sample rate. This may seem at odds with the traditional concept of a sampler, that of a "tape recorder" which transposes pitch up by playing the tape faster and down by playing it slower. The concept of changing tape speed is the analog equivalent of changing the playback sample rate in a sampler. It turns out that there are a number of disadvantages to changing the sample rate and a number of advantages to maintaining a constant sample rate. This means, however, that we are talking about raising or lowering the pitch without "changing the tape speed" -- how do we do that? Well, we can't do that with a tape recorder, but we can with digital electronics. Essentially what happens is, we transpose up in pitch by skipping over samples -- to transpose up an octave, we play back every other sample. To transpose down in pitch, we play samples more than once -- to transpose down an octave, we play back each sample twice.

The disadvantage of a variable output sample rate system is complexity, expense and the limitations it places on the system (it is difficult to perform independent digital processing on each voice and the number of voices and transposition range are limited). It is virtually assured that all systems in the future will be fixed-rate systems in order to allow digital keyboards, digital mixing consoles and digital recorders to talk to each other directly in digital form.

There are disadvantages to fixed-rate systems as well. As you might imagine, skipping over samples can lead to distortion when transposing sounds up in pitch (the less of the available samples you use, the less accurate the final waveform becomes). In addition, what effectively happens is, the sampled waveform in memory is *re-sampled* as it is played back. It is quite possible that waveform data in memory may not alias when all of the samples are played back in order, but may still contain harmonics which alias when transposed by this re-sampling playback process. That's because we are violating the rules of sampling. When the sound was originally sampled, we made sure that nothing would alias, but now we are not playing back all of those samples, instead we're making up a new waveform which may indeed alias. This is the reason that sounds from the DOC I tend to get "dirtier" when transposing up in pitch.

To eliminate these problems, what we would really like to have is a continuous waveform in memory so that we can resample it any way we like for playback. This isn't possible since, by definition, a digital sample can't be continuous -- if it was, it would be neither digital, nor a sample! We can approximate a continuous waveform by *oversampling* a sound, that is, sampling the input at a much higher rate than is normally needed to avoid aliasing. This insures there are more than enough closely-spaced samples so when we start skipping them, we still produce a fairly accurate waveform. This technique has been used successfully on many Mirage sounds, but it wastes memory since it requires many more samples than needed by other samplers. What is needed is a method of taking the normal amount of samples in memory and approximating a continuous waveform.

Oversampling and Interpolation

Using a process called *Linear Interpolation*, the DOC II eliminates the aliasing problems of the original Mirage and even goes one better on other samplers. The interpolation process generates new samples based on a weighted average of existing samples. When you transpose up in pitch, the DOC II *makes up* samples which correspond to the samples that should be produced at that pitch. The DOC I simply put out whatever sample was in memory at a given location, even if the desired value was closer to the sample in the next location. The interpolator can actually synthesize up to 512 different sample values between each "real" sample in memory -- effectively performing oversampling in real-time, without wasting any memory.

When most samplers transpose down in pitch, the sound gets "grainy" as downward transposition effectively reduces the output sample rate. Again, the DOC II maintains a fixed output sample rate. When transposing down, the interpolator synthesizes samples between the real samples, filling the gaps. This eliminates the graininess. The interpolation process allows the DOC II to sound as good as a variable rate sampler when transposing up and better when transposing down. At the same time, the DOC II is less expensive, has more voices, a wider transposition range and allows other digital signal processing such as *Digital Filtering.*

Digital Filters

Each voice of the DOC II has its own multi-mode dynamic digital filter. Other samplers use either conventional synthesizer VCFs (like the Curtis chips we use in our other products) or simply don't provide dynamic filtering. It is difficult to reproduce instrument sounds without dynamic filtering which, for example, allows the brightness of a sound to vary with velocity. Conventional VCFs have their own problems (as we all know!). VCFs are noisy, they distort and pop and are marginally reliable. On top of that, they're expensive and have a lot of support components. The digital filters in the DOC II are 100% reliable and consistent (if the chip works, the filters work!). They are very low in distortion and noise because they are actually digital equations which model the operation of an analog filter. Filter quality is a function of the accuracy of the calculations, which are performed on the DOC II using 24 bits of resolution. This is far superior to any analog filter. On top of that, the filters have selectable high and low pass modes for more flexibility compared to other samplers. The only analog filters in the EPS are the two fixed-frequency filters on the stereo outputs which smooth the output waveform and filter out the sample rate clock. They have none of the problems associated with VCFs.

Miscellaneous

As in the Mirage, the DOC II chip generates all of the timing signals and refresh functions required by the dynamic RAMs. It also contains the input A/D converter for sampling and the output D/A converters for playback. The output actually consists of the top 13 bits of the digital filter calculations, multiplied by 12 bits of volume information (16 times better than the DOC I) which yields a 25-bit floating-point output. The floating point conversion improves the output noise over a linear output, as the signal-to-noise ratio remains constant when sounds decay. In conventional converters, the lower the signal gets, the worse the signal-to-noise ratio becomes. Finally, the DOC II controls the interface between the synchronous Sound bus and the asynchronous 68000 bus, which brings us to the....

68000 MICROPROCESSOR

The 68000 is one of the most powerful microprocessors available. It is the same chip used in the Macintosh, Atari ST and Amiga computers, as well as a number of powerful CAD workstations. The EPS uses a 10 MHz clock, which is nearly 40% faster than the Macintosh. The 68000 is similar to the 6809 processor used in our other systems, but has a number of key differences. First, the 68000 has a 16-bit data bus (as compared to the 8-bits of the 6809) and a 24-bit address bus (as compared to 16-bits on the 6809). This means there are a lot more lines to get shorted out or screwed up.

Asynchronous Bus

Unlike the 6809, which is a synchronous processor, the 68000 uses an asynchronous bus. What does that mean? The 6809 has a clock signal (the "E" clock) to which all operations are synchronized. On the rise of E, the 6809's addresses are stable and on the fall of E, data is transferred. This is a simple system, but it has limitations. If a device being accessed is too slow to provide data by the fall of E, the system will crash. That means all devices on the bus must be fast enough to keep up. It also means that using devices which are faster than the bus speed is a waste. Even if the data is available sooner, the 6809 still waits until the fall of the E clock to accept the data.

The 68000, on the other hand, has a control signal called DTACK (*Data Transfer ACKnowledge*), which allows the bus to run at whatever speed a given device needs. What happens is, the 68000 initiates a read or write of a device, then stops and waits until the DTACK line is brought low. The device being accessed will bring DTACK low when it has accepted the data from the processor or when it has valid data to give the processor. This technique allows slow and fast devices to be mixed on a single bus. Slow (therefore inexpensive) devices can be used where possible without degrading the speed of faster components. This

allows system performance and cost to be optimized. It can also lead to a system crash should the 68000 ever access a block of memory which doesn't have a device associated with it. In that case, the 68000 would halt forever, waiting for a DTACK which will never happen. Under normal circumstances, the 68000 will never access illegal locations, however common board problem (shorts, opens, bad chips, etc.) could cause this state, so it may be a common occurrence during debugging.

Upper/Lower Bytes and Words

The address bus of the 68000 is a bit (no pun intended) more complicated than it seems. Although addresses are 24 bits wide, only the high-order 23 bits come out of the chip. The least significant bit, A0, is used internally. Although the 68000 can address memory as 16-bit words, it can also individually access either byte of a word. The 68000 seperates a word into *Upper* and *Lower* bytes. When accessing bytes, if A0 is low (an even address), the 68000 is selecting the upper byte of the word. If A0 is high (an odd address), the 68000 is selecting the lower byte of the word.

In order to control this selection process, the 68000 has two control lines called UDS (*Upper Data Strobe*) and LDS (*Lower Data Strobe*). When accessing an upper byte (an even address), UDS will be low. When accessing a lower byte (an odd address), LDS will be low. When accessing a word (an even address), both UDS and LDS will be low. Words can only be accessed on even addresses. Accessing a word on an odd address doesn't make sense as this would be an attempt to access the lower byte of one word and the upper byte of the next word -- two bytes which are unrelated. Any attempt to perform a word access on an odd address will generate a system error which will put the system into an endless loop. Again, under normal circumstances, this will never happen, but while debugging boards, it may crop up. In the event this occurs, you will see the 68000 executing a loop, but none of the normal operations will occur (display messages, etc.).

Interrupts

Interrupts on the 68000 are more complicated than on the 6809, but are actually easier to debug. The 68000 uses a system called *Vectored Priority Interrupts*. "Priority" means there are different levels of interrupts, each of which have different importance. On the 6809, all interrupts were common and any interrupt could conceivably interrupt the processing of another. On the 68000, each interrupt (up to 7) has a priority level. Higher priority interrupts can interrupt lower priority interrupts, but lower priority interrupts will be ignored until all higher priority interrupts are finished being serviced. This insures that priority operations will not be disturbed (for example, the highest priority on the EPS is MIDI interrupts. Missing a MIDI key-up message would be a disaster as a note would hang forever). Only three levels are used on the EPS; level 1 is the DOC chip, level 2 is the DMAC and level 3 (the highest priority) is the DUART. Levels 4 through 7 are not used and level 0 indicates that no interrupts are pending.

"Vectored" means that each interrupt directly informs the processor where to find the software routine which services it. On the 6809, when an interrupt was received, the processor had to examine each device capable of generating an interrupt to determine who needed service, then it had to figure out where the proper service routine was. This added a lot of time to interrupt processing, bogging the system down. When the 68000 receives an interrupt, it generates an *Interrupt Acknowledge* (IACK) to the interrupting device. In response, the device puts out a code on the data bus which tells the 68000 where the appropriate service routine in memory is, whereupon the processor immediately goes there. This happens very quickly with virtually no software overhead.

This vectoring process assumes that the device has the ability to generate an interrupt vector, which isn't true in all cases. For example, both the DUART and DMAC produce vectors, since they were designed specifically to work with the 68000. On the other hand, the DOC does not generate a vector. The 68000 has a special *Autovector* mode for such occasions. If, in response to an IACK from the processor, the VPA input to the 68000 is pulled low, the processor will generate its own vector appropriate to the interrupt level being serviced.

To control all of the interrupt functions, the EPS uses a special PLA called the ICU (*Interrupt Control Unit*). This chip detects when interrupts occur and encodes the Interrupt Priority Level on the 68000 inputs IPL0 and IPL1. It then decodes the Interrupt acknowledge state of the processor and generates the IACK lines for

the DMAC or DUART, or the VPA line for the DOC. The ICU along with the two other PLA's, the MMU (*Memory Management Unit*) which decodes the memory map of the system and the BCU (*Bus Control Unit*) which generates all of the system control signals will be described in more detail later.

Because each interrupt line can be viewed individually, it is often easier to debug 68000 interrupt problems than with the 6809. When the 6809 crashes due to interrupt problems, you never really know which device is causing it, as all of the interrupt lines are common. During normal operation of the EPS, the only active interrupt is from the DUART, no other interrupts should be going low. Interrupts from the DMAC only occur during DMA disk accesses or when sampling, and interrupts from the DOC only occur during sampling. If any of these lines are low (or moving) when they're not supposed to, there's something wrong and the system will act very strangely. Additionally, both IPL2 and BERR must be high for proper operation.

SOUND RAM BUS SYNCHRONIZATION

The DOC chip continuously accesses Sound RAM at a fixed rate which is proportional to the output sample rate. The DOC is in charge since the most important thing is to make good sound -- nothing can interfere with the DOC. Accessing memory at a fixed rate implies a synchronous system, which is the case. The DOC chip and the Sound RAM form a completely independent synchronous subsystem in the EPS.

This subsystem is separated from the asynchronous 68000 bus by the 74LS245 data buffers, 74LS244 address buffer and the 74LS253 address multiplexors. The enables of the 74LS245's and 74LS244 are controlled by the BCU, while the direction is controlled by the 68000 R/W line. The 74LS253's serve to select between DOC or processor addresses and to multiplex the high and low order addresses together, as required by the dynamic RAMs. They are controlled directly by the DOC chip with the signals DBUS (which selects DOC addresses when low or processor addresses when high) and AMUX (which selects the low order part of the address when high and the high order address when low).

In order for the 68000 to access the Sound RAM, it requests an access via the SRAM line, which goes into the DOC. The 68000 is held off from the Sound RAM until DOC is finished getting samples. When DOC is no longer accessing the RAM, it allows the 68000 access and gives the 68000 a DTACK. Each time the 68000 tries to access the Sound RAM, it will be held off via the DTACK line, until there is an opening. The same process is used when the processor tries to access the DOC chip registers. The processor requests an access of the DOC via the DOC Chip Select line. The DBUS signal generated by the DOC chip determines who has access to the Sound RAM. When it is low, the DOC chip is accessing memory and the processor must be held off. When DBUS is high, the processor can get into the Sound RAM or access the DOC chip registers.

TIMING

The timing generators for the system are very simple. There are two self-contained oscillator modules which eliminate the inverters, resistors, capacitors and crystals which we have used in the past, greatly improving consistency and reliability.

The 10 MHz module drives the 68000, DMAC and DOC chip. In addition, Flip-Flop U34b is used as a divider to generate the 5 MHz master clock required by the DUART. The 68000 has an output clock called "E" which is a divide-by-ten of its input clock. With 10 Mhz going in, the E clock comes out at a frequency of 1 Mhz. This 1 Mhz is fed into the DUART to set the communications rate of the KPC port. U34a is used to divide E by two for a 500 KHz clock, which is fed into the DUART to set the rate of the MIDI port. The 8 MHz oscillator drives the FDC only, as this frequency is required for use with standard floppy disks.

There is also a separate timing generator for the switched capacitor Input Sampling Filter used in the analog section. As the EPS offers a number of different input sampling rates, it is desirable for the Input filter to track these various rates. The clock for the filter (FCLK) is generated by a programmable divider. The frequency is controlled by the settings of CA0, CA1, CA2 and AN1. These signals are normally used for playback functions, but during sampling are inactive, allowing their use in this divider circuit.

The circuit consists of a 74LS161A programmable counter (U37) and a 74LS74 flip-flop (U39). U39a is used to capture the carry output of the counter and reload the counter with the desired starting count (set by CA0-CA2 and AN1). Different starting counts produce different output frequencies. U39b divides the carry output by two to generate a square wave clock as required by the filter. The MUTE line allows the divider to operate when the output is muted (during sampling) but forces a constant clock when the output is not muted (during normal playback) by holding the counter in the load state. This prevents FCLK from changing randomly as the control lines (CA0, etc.) move.

RESET

The 555-based Power-On-Reset circuit is the same as used on the Mirage, but with a longer time constant (the RESET output will be low for approximately 900 mS). The 68000 requires both it's RESET and HALT inputs to be pulled low at power-on. Also, the 68000 is capable of putting a RESET signal out on this line under certain condtions. In normal operation, this will never happen, but don't be surprised if you see this line moving when debugging bad boards -- it's most likely not the fault of the 555.

PLAs

ICU

Portions of the ICU were previously described. This section will detail the inputs and outputs as programmed into the PLA.

ICU Inputs:

AS - Address Strobe. This signal comes from the 68000 and goes low whenever a valid address is present on the processor address bus and when a valid code is present on the processor status lines (FC0, FC1 and FC2). The ICU outputs are gated with this signal to insure that none of the output lines glitch while new addresses are stabilizing.

FC0, FC1, FC2 - These signals are the processor status lines which indicate what state the processor is in at any given time. There are eight possible states, but the only two we are concerned with are the Interrupt Acknowledge state (HHH-when the processor is acknowledging an interrupt) and the Supervisor Program state (HHL-which the processor enters at RESET time). When the interrupt acknowledge state occurs, the ICU generates the appropriate Interrupt acknowledge by decoding the processor address lines to determine the interrupt level. The occurence of the supervisor program state is used to determine that a RESET has occurred and that the Reset vector should be fetched from EPROM (more on this under SUPER).

A1, A2 - These are the low order address lines from the processor. During an Interrupt Acknowledge cycle, the interrupt level being serviced is encoded on these lines and must be decoded by the ICU to generate the appropriate interrupt acknowledge signal.

DUIRQ - This is the interrupt request from the DUART. The DUART will bring this line low when it has an interrupt that needs service (timer, MIDI, message from the KPC, etc.).

DMIRQ - This is the interrupt request from the DMA controller. The DMAC will bring this line low when it has an interrupt that needs service (interrupt from the FDC, DMA data transfer complete, etc.).

DOCIRQ - This is the interrupt request from the DOC chip. The DOC will bring this line low when it has an interrupt that needs service (oscillator interrupt, etc.). This line normally goes low only during sampling. If a steady stream of interrupts is not seen on this line during sampling, something is wrong.

DT - This is the DT line from the MMU (more on this under the MMU). The ICU is used to convert this signal into an open-collector output for use as the DTACK line. This function really has nothing to do with interrupts, it was simply a matter of the ICU chip having a spare input and output which weren't being used.

ICU Outputs:

IPL1, IPL0 - These two lines go to the interrupt inputs of the 68000, to inform the processor that an interrupt has occured. Each of the three interrupts produces it's own code (HH=no interrupt, HL=DOC interrupt, LH=DMAC interrupt, LL=DUART interrupt). The ICU also controls the priority of the interrupts, allowing a higher level interrupt to pass it's code along, or preventing a low level interrupt from passing it's code along in the presence of a higher level interrupt. If more than one interrupt input is low, the output code will be that of the highest priority interrupt.

DUIAK - This is the interrupt acknowledge line to the DUART. When the processor acknowledges an interrupt from the DUART, this line is brought low, causing the DUART to place it's interrupt vector on the data bus.

DMIACK - This is the interrupt acknowledge line to the DMAC. When the processor acknowledges an interrupt from the DMAC, this line is brought low, causing the DMAC to place it's interrupt vector on the data bus.

VPA - This line goes to the VPA input of the 68000. When the processor acknowledges an interrupt from the DOC chip, this line is brought low to cause the processor to generate an autovector.

IACK - This line goes to the MMU. Whenever the processor acknowledges an interrupt, this line goes low to prevent the MMU from selecting any device whose output data might fight with the vector from the interrupting device.

SUPER - This line also goes to the MMU. Whenever the processor gets RESET, this line goes low to inform the MMU that the Reset vector from the EPROM should be placed on the bus.

DTACK - This line is connected to all other DTACK outputs and goes into the 68000. The output is open collector (as are the DTACK outputs on all other devices) allowing DTACK outputs to be "wire-OR'd" together. Essentially, this output does nothing more than turn the DT line from the MMU into an open-collector output.

The basic sequence of events for an interrupt is this: One of the devices (DUART, DMAC, DOC) needs an interrupt serviced, so it pulls it's interrupt line low. The ICU places the code appropriate to that interrupt on IPL1 and IPL0. The processor detects the interrupt and goes into an interrupt acknowledge state, placing the interrupt acknowledge code on FC2, FC1 and FC0 and placing the encoded interrupt level on A2 and A1. On the fall of AS, the ICU detects the interrupt acknowledge state, pulls IACK low, decodes the interrupt level and pulls the appropriate interrupt acknowledge line (or VPA) low. If a higher priority interrupt occurs, the ICU will pass it's code along to IPL1 and IPL0, causing a new interrupt. If a lower priority interrupt occurs, the ICU will ignore it until the higher priority interrupt line again goes high, at which point the lower priority interrupt will be allowed to happen.

MMU

The MMU (*Memory Management Unit*) is used to decode the address space of the microprocessor and generate the chip selects for the various devices. It is also used to generate a DTACK signal for those devices which don't provide a DTACK output.

MMU Inputs:

SUPER - This is the signal from the ICU which indicates that the processor has entered a RESET state. It is used to override the normal chip select decoding and force the MMU to select the EPROMs so that the Reset vector is placed on the bus.

IACK - This is the signal from the ICU which indicates that the processor is acknowledging an interrupt. It is used to overide the normal chip select decoding and prevent the MMU from selecting any device which could interfere with the interrupt vector fetch.

A23, A22, A21 - These are the high order addresses from the processor. The MMU decodes these addresses into the various chip select outputs.

AS - Again, this is the signal from the processor which indicates that addresses are valid. All chip select outputs are gated with this signal to prevent glitches.

DBUS - This is the signal from the DOC which indicates whether the DOC or the processor has access to the Sound RAM. It appears as a free-running 1.25 MHz square wave and is used to determine whether processor addresses or DOC addresses are used to select the Sound RAM. When DBUS is low, the DOC is accessing Sound RAM, so the DOC address (DA18L) is used to select the appropriate RAM bank. When DBUS is high, the processor has access to the Sound RAM, so the processor addresses are used to select the RAM.

DA18L - This is DOC address A18, which has been latched by flip-flop U30b. Like the DMAC, the DOC chip shares addresses and data on the same pins. The flip-flop serves the same purpose as the latches used with the DMAC, it captures the address information and holds it valid for the entire cycle. The other DOC address lines do not need to be latched as they only go to the Dynamic RAMs, which have built-in address latches. This line is used to determine whether the DOC is accessing the internal or external Sound RAM banks. When it is low and DBUS is low, the DOC is accessing the internal Sound RAM. When it is high and DBUS is low, the DOC is accessing the expansion RAM. When DBUS is high, the state of this line is irrelevant.

1M - This is an input from the memory expansion port. Normally this line is always high. It is brought low only when the 4X memory expander is installed. When low, it overides the normal Sound RAM decoding and forces the DOC to only access the expansion RAM, regardless of the state of DA18L. The processor then has the internal memory all to itself for use with the sequencer (the internal memory would no longer be Sound RAM, as the DOC cannot access it).

CAS - This is the Column Address Strobe from the DOC chip. It is used as a reference for generating the Dynamic RAM chip selects (CAS0 and CAS1).

FDCDLY - This is the Floppy Disk Controller chip select, delayed by one 10 MHz clock period using flip-flop U30a. The FDC does not have a DTACK output and neither do the RAMs or EPROMs. The MMU must generate a DTACK output for these devices. The RAMs and EPROMs are fast enough to generate DTACK directly from their chip selects, but the FDC is too slow for this. DTACK for the FDC is generated from this delayed FDC chip select, which holds off DTACK slightly to give the FDC enough time.

MMU Outputs:

OSROM - This is the chip select for the Operating System EPROMS and goes low whenever the processor is accessing the memory range $C00000 - $DFFFFF, or when a RESET condition occurs.

OSRAM - This is the chip select for the Operating System Static RAMs and goes low whenever the processor is accessing $000000 - $1FFFFF or $E00000 - $FFFFFF. It is mapped into both locations because of certain software efficiencies that are possible with the 68000. During Supervisor Program state, the RAMs are forced off so that the EPROMs can be selected.

I/O - This is the chip select for the I/O area and goes low when the processor accesses $200000 - $3FFFFF. The I/O space is further decoded by the 74LS138 (U32), which generates the separate low-active chip selects for each I/O device. U32 breaks the I/O space down into these blocks:

$200000	-	$23FFFF	= DOC
$240000	-	$27FFFF	= DMAC
$280001	-	$2BFFFF	= DUART (odd bytes only)
$2C0001	-	$2FFFFF	= FDC (odd bytes only)
$300000	-	$33FFFF	= Expansion I/O
$340000	-	$37FFFF	= Expansion I/O
$380000	-	$3BFFFF	= Expansion I/O
$3C0000	-	$3FFFFF	= Unused

SRAM - This is the chip select request for the Sound RAM which goes low whenever the processor accesses $400000 - $7FFFFF. This line is used to inform the DOC that the processor wishes to access the Sound RAM. The DOC handles all arbitration from that point.

CAS0 - This is the chip select for the internal Sound RAM bank. It goes low during a processor access of $400000 - $5FFFFF when DBUS is high. It will go low during a DOC access (DBUS low) if DA18L is low and the 4X expander is not installed. This signal is gated by the MMU with the DOC CAS reference for proper timing.

CAS1 - This is the chip select for the external Sound RAM bank. It goes low during a processor access of $600000 - $7FFFFF when DBUS is high. It will go low during a DOC access (DBUS low) if DA18L is high and the 4X expander is not installed. If the 4X expander is installed (1M low), CAS1 will go low on every DOC access and CAS0 will not go low on any DOC access. This signal is gated by the MMU with the DOC CAS reference for proper timing.

DT - This is the DTACK signal generated by the MMU for those devices which don't have a DTACK output (OSROM, OSRAM and FDC). This line will only go low when one of these devices is being accesses. Because this output is not open collector, as required by the 68000 DTACK line, it is sent to the ICU, where it is converted to an open-collector output.

BCU

The BCU (*Bus Control Unit*) generates all of the control signals needed by the system to select upper and lower bytes and communicate with the synchronous Sound bus.

BCU Inputs:

UDS - This is the Upper Data Strobe from the 68000. It goes low when the 68000 is accessing a word or an upper byte of a word.

LDS - This is the Lower Data Strobe from the 68000. It goes low when the 68000 is accessing a word or a lower byte of a word.

R/W - This is the Read/Write line from the 68000. It is high on a read and goes low on a write.

SRAM - This is the SRAM signal from the MMU which goes low when the processor wants to access the Sound RAM.

DBUS - Again, the DOC signal which tells when the DOC has control of the Sound bus and when it is free.

DOC - This is the chip select to the DOC chip which goes low when the processor wants to access the DOC chip.

OWN - This is a control signal from the DMAC which goes low when the DMAC has control of the bus.

DBEN - This is a control signal from the DMAC which goes low when the DMAC has control of the bus and is actually transferring data. DBEN is used to delay the chip selects and output enables when OWN is low, to prevent data from appearing on the bus until the DMAC is ready for it. When OWN is high (i.e. during a non-DMA access), this delay is not necessary, so DBEN is ignored.

BCU Outputs:

UWE - This is the upper write line, which goes low when the processor is writing a word or an upper byte.

LWE - This is the lower write line, which goes low when the processor is writing a word or a lower byte.

URD - This is the upper read line, which goes low when the processor is reading a word or an upper byte.

LRD - This is the lower read line, which goes low when the processor is reading a word or a lower byte.

The preceding control lines are used to select which of the Static RAMs are being written to and which of the Static RAMs or EPROMs are being read from.

U245EN - This is the enable line for the 74LS245 buffer on the upper byte of the Sound RAM data bus. It goes low when the processor is accessing a word or upper byte of Sound RAM, or when accessing the DOC. This signal is gated with DBUS so the buffer cannot be enabled during a DOC access, which would cause a bus fight. It is also used to enable the 74LS244 buffer allowing the processor's addresses to go to the DOC chip when accessing the DOC registers.

L245EN - This is the enable line for the 74LS245 buffer on the lower byte of the Sound RAM data bus. It will go low when the processor is accessing a word or lower byte of the Sound RAM, or when accessing the DOC chip. This signal is gated with DBUS so that the buffer can never be enabled during a DOC access, which would cause a bus fight.

244EN - This is the enable line for the 74LS244 buffer which pulls the lower 3 bits of the Sound RAM bus low when the processor is reading from the Sound RAM. Since there are only 13 bits of Sound RAM, this insures the lower 3 bits, which don't really exist, are always read as lows. The buffer will not turn on when accessing the DOC chip, as the DOC registers are a full 16 bits wide, nor will it turn on during a write to the Sound RAM, since there are no RAM chips attached to the lower 3 bits to be written to.

UGWE - This is the upper write line to the Sound RAM. It is gated with DBUS to insure that no writes can occur during a DOC access. It will only go low when DBUS is high and the processor is writing an upper byte or word of Sound RAM.

LGWE - This is the lower write line to the Sound RAM. It is gated with DBUS to insure that no writes can occur during a DOC access. It will only go low when DBUS is high and the processor is writing a lower byte or word of Sound RAM.

DS - This is a signal which OR's together the upper and lower data strobes. It is used to insure that the I/O devices can only be selected when the processor is ready to transfer data (by enabling the 74LS138 decoder). It is necessary to OR the signals together since some I/O registers are only on the upper data bus, some are only on the lower and some are on both.

When writing or reading words from the Sound RAM or DOC, both 74LS245 buffers will be enabled. The DOC chip can only be accessed as words as all of it's registers are 16-bit. When writing a lower byte to Sound RAM, the lower buffer will be enabled and the lower write line will go low. The upper write line will remain high, which means that the upper byte of RAM chips is being read from whenever the lower byte is being written to. This doesn't cause any problems as the upper data buffer is off, preventing any bus contention. The same situation exists with lower bytes when writing upper bytes.

This is a somewhat unconventional approach to the 68000 bus. Normally, a common write line is used with seperate chip selects being generated for the upper and lower bytes. A 68000 system running at 10 MHz, however, simply doesn't allow enough write access time for reasonable speed memory chips. This solution allows a 10 MHz system to function with reasonable cost RAM chips.

MISCELLANEOUS

Bus Termination

All address lines and control lines have 3.3 KΩ pull-up resistors as there are times when control is being passed to and from the DMAC that these lines go tri-state. The pull-ups prevent these lines from floating and causing erroneous bus states.

Connectors

The Power connector is our standard Molex type which provides unregulated voltage for the analog circuitry (+VA, -VA and AGND), unregulated voltage for the Poly-pressure Keyboard (+VU), two separate regulated +5 volt outputs (+5D and +5M) and ground returns (DGND). The source for +5D is also brought over to the analog section (+5A) by a seperate wire, eliminating any digital noise which +5D may pick up on the main board.

The 7-pin Molex Display connector passes the communication signals from the KPC board (MISO, MOSI and BSCK) to the Display Board. It also supplies the Display Board clock (CLK) and brings the Value and Volume slider output voltages (VALV and VOLV) down to the main board.

The Keyboard connector is a 20-pin post header which carries power to the KPC board along with the display communication signals and the signals which allow the main board to communicate with the KPC board (SERIN and SEROUT). In addition, the footswitch signals are sent over from the main board to be encoded by the KPC processor.

The Memory Expansion connector is our standard 44-pin connector and supplies all of the signals necessary to interface to the expansion Dynamic RAM cartridges. It also supplies a number of signals which will be used with future expansion modules (such as the SCSI port).

The Auxilliary Expansion connector is a 34-pin post header which brings out every signal that could conceivably be used with an expansion device. Some of these signals are used with the SCSI port, while others are there for possible future expanders.

The Disk connector is our standard disk connector. Since this connector is also a 34-pin post header, don't confuse it with the Aux Expansion connector.

It is impossible to go into every detail of a system as complex as this which took over a year to develop. This overview is designed to give you some insight into the operation of the EPS so that you can develop your own tips and skills in debugging the boards. As more tricks are learned from experience, we can add to the debugging procedure.

An Overview of the EPS Main Board Analog Section

GENERAL

The analog section of the EPS consists of all of the support circuitry the DOC II chip needs to operate, along with the sampling circuitry, output circuitry and analog input circuitry (wheels, sliders, etc.). The design is similar to that used in the Mirage, however the DOC II demands superior circuitry in order to realize the sonic benefits of it's higher resolution. In other words, we didn't need to be too critical with the DOC I because the benefits of better circuitry would not have been audible in an 8-bit system. The 13-bit precision of the DOC II is 32 times better than DOC I, so all other circuitry must be up to this level of performance.

By the way, there are no sampling keyboards under $60,000 which provide this level of sampling performance. Virtually all of our competitors are 12-bit systems and the so-called "16-bit" CASIO FZ-1 does not *sample* with 16-bit resolution. The FZ-1 plays back 16-bits, but input sampling quality measures worse than 12 bits. That means we have *twice* the resolution of our competitors (remember, each additional bit doubles the range--we have 8192 steps versus 4096 for 12-bit systems).

Of course one *big* difference between the EPS and our older systems is that there are no voltage-controlled filters (i.e. Curtis chips) and since there are no control voltages, there are also no SSM 2300 chips. Instead, all filtering, volume control and panning control are digital. The circuitry can be broken down into a number of functional blocks which are *Input Circuitry*, *Output Circuitry* and *Support Circuitry*.

SUPPORT CIRCUITRY

The Support Circuitry consists of the *Power Supply, Reference and Volume* and *Analog Timing* blocks shown on the schematic.

Power Supply

The analog section requires regulated +12 VDC, -12VDC, +5 VDC and -5 VDC. The 12 volt supplies are used to power the op-amps while the 5 volt supplies are used to power the various CMOS chips. Unlike our other products, the 12 volt supplies are generated on the main board rather than on the power supply itself. This is to insure the lowest possible noise on the analog supplies. Unregulated DC voltage is brought over to the main board where Q1 and Q2 turn it into regulated 12 volts.

The CMOS chips draw very little current and are relatively immune to noise on their supply lines. The +5 voltage is simply the +5 digital supply, however rather than tapping it off the main board, it is brought over from the power supply on its own wire. This eliminates any digital noise which is produced on the main board. The -5 voltage is generated by a resistive voltage divider from the -12 volt supply. It is buffered by U49a to create a low-impedance output.

The grounding scheme of the EPS is quite complex in order to achieve minimum noise levels. Grounds for each analog circuit block (input, output, etc.) are kept separate and commoned at a single point, which is fed back to the power supply as a separate analog ground.

Reference and Volume

Like the DOC I, the DOC II requires an external voltage reference for it's internal convertors. The DOC I simply used the -5 volt supply as a reference, however for the DOC II a higher output level was desired to improve the signal-to-noise ratio. The maximum reference the DOC II can handle is +9.5 volts. A resistive voltage divider off of the +12 volt supply, buffered by U49d, is used to generate the +8.75 volt reference.

Operation of the Volume D/A converter in the DOC II is identical to the DOC I. It requires an op-amp inverter (U50d) to convert the current output of the Volume D/A into a voltage. The only difference is the resulting volume output pulses are negative going because a positive reference is used (on DOC I, a negative reference was used which produced positive volume pulses).

The Volume feedback resistor is built into the DOC II. This resistor, together with the Waveform feedback resistors which are also built-in, eliminates the "hot DOC" problem that often occured with DOC I. With all of the critical resistors contained on the chip, any variations in resistance automatically cancels out, resulting in consistent output levels from chip to chip without trimming.

The resolution of the DOC II is such that even minute variations in internal resistance can add up to an error when the most significant bit of the output converter switches in. This error can result in distortion on low level signals (or when a sample is fading out). In order to compensate for this error, an MSB Adjustment input is provided which can be set by Pot 1 (*OUTPUT MSB ADJUST*).

Analog Timing

The analog timing block consists of support circuitry for the other functional blocks. These various circuits are more appropriately described with their associated blocks, so they will be covered seperately. A description of the Output Expander circuitry can be found in the *MISCELLANEOUS* section at the end.

OUTPUT CIRCUITRY

The output circuitry consists of the *Waveform Converters and Diff. Amp, Panner and Deglitcher* and *Output Filter* blocks shown on the schematic.

Waveform Converters and Diff. Amp

The DOC chips use a *Floating Point* output conversion in which volume control of waveforms is performed in the analog domain. Most digital systems process volume digitally before the signal is sent out the final D/A converter--the actual digital data is "scaled-down". This has the undesirable characteristic of degrading the signal-to-noise performance of the output when sounds get quieter since less of the dynamic range of the output D/A is used.

The DOC chips modulate the reference voltage of the output Waveform D/A converter to control volume. In order to make a sound quieter, the reference is reduced, but the waveform data continues to use the full range of the D/A. Rather than the S/N getting worse as sounds decay, the DOC chips maintain a constant S/N whether the sound is at full volume or nearly off. This is why the DOC II, with 13 bits of waveform data and 12 bits of volume control, provides 25 bits of output dynamic range. This is a theoretical dynamic range of 150 dB, which is far beyond the range of anything! Of course there are real-world limitations--there are no op-amps capable of such performance, so only 96 dB of dynamic range is claimed for the EPS, the present limit of reasonable circuitry.

Some claim we are cheating because, in fact, the best signal-to-*quantizing* noise ratio that can be achieved with a 13-bit system is 78 dB. It is true that if the EPS is used purely as a digital recorder, without making use of the envelopes or digital filtering, the limit is 78 dB (which is still excellent). If, however, a sound is sampled with the assumption that the envelopes will be used to control the dynamics of the sound, 96 dB performance is easily achieved and the factory sounds are sampled with this in mind.

The Volume output converter, described in the previous section, actually generates the reference voltage for the Waveform D/A. If you look at the output of the Volume converter while playing notes, you will see a group of narrow pulses that follow the volume envelope of the sound. Each pulse represents the volume of a particular oscillator at that point in time. Note that it doesn't matter the volume pulses are negative-going, all this does is make the Waveform outputs (which are inverted) positive going. The Diff. Amp which combines the Waveform outputs is unaffected by this.

As with the Volume converter, operation of the Waveform D/A converter is identical to the DOC I. The D/A is brought out as two differential output currents (SIG+ and SIG-). These current outputs are converted into voltages by inverting op-amps U65b and U64b. Both of these voltages are positive going signals which correspond to part of the waveform and track the Volume output. The voltage generated by SIG+ represents the positive half of the waveform, while the voltage generated by SIG- represents the negative half. Again, because the waveform feedback resistors are built into DOC II, there is no variation in output level from chip to chip. The two signals are combined into a single bipolar (positive and negative) waveform by Diff. Amp (short

for *Differential Amplifier*--a fancy name for an op-amp that subtracts two signals) U64a. The output of the Diff. Amp is the final waveform, whose amplitude is a function of the Volume signal.

Panner and Deglitcher

Stereo panning on the EPS is performed differently than on any of our previous products. On the Mirage DSK, each voice was hard panned to a different spot in the stereo field. On the ESQ/SQ-80, panning was accomplished in the Curtis chips by a voltage-controlled mixer. On the EPS, stereo pan is accomplished using a technique which also allows programmable polyphonic solo outputs. Like the DOC I, each oscillator in the DOC II has a 4-bit channel assignment code (CA0-CA3). This allows an oscillator to be sent to any of 16 locations depending on its code. These locations could be 16 solo outputs or, in the case of the EPS, 8 stereo field positions or one of 8 solo outputs.

The final waveform output from the Diff. Amp is sent to three resistors (R95, R96 and R97) which are weighted to produce different amounts of signal into the three CMOS switches of the 4053 Panner (U55). Depending on the channel assign code (CA0, CA1 and CA2) the signal from each resistor is sent to either the left bus (U50a) or right bus (U50b). The various combinations of the three resistor values and which bus the signal from each resistor is sent to results in 8 different pan postions. In order to allow for the optional solo output expander, CA3 is sent to the *Inhibit* input of U55. Whenever CA3 for a particular oscillator is low, the Panner is enabled and the oscillator is sent to the appropriate position in the stereo outputs as selected by CA0, CA1 and CA2. Whenever CA3 for a particular oscillator is high, the Panner is disabled so the oscillator will not appear at the stereo outputs. Instead, the oscillator is sent to the Output Expander where CA0, CA1 and CA2 select which of the 8 solo outputs the oscillator will appear at (further discussion of the Output Expander follows).

The Deglitcher follows the Panner. As with the DOC I, DOC II has a signal called *CSTB* (Channel Strobe) which goes low when the analog output of an oscillator is valid. There is a lot going on when an oscillator is active. The Volume and Waveform D/A's must stabilize at their correct values, the Panner switches must then set up and, finally, the Panner outputs must stabilize. During this period the Panner outputs must be ignored because they are not valid. Once the outputs of the Panner have stabilized, CSTB goes low. This signal is used to generate a control line (*PAN CSTB*), which drives the *Inhibit* input of the 4053 Deglitcher (U54). When enabled by PAN CSTB, the switches in U54 allow the Panner outputs to pass to the final output. The switches do nothing more than prevent the Panner signals from getting to the output until they are valid.

The Deglitcher also serves another purpose. The *MUTE* control line from the digital section is used to select whether the DOC output or the Sampling input is sent to the audio output. When the output is *not* muted (during normal playing), the signal at the sampling input (*FILT*) is off. When the output *is* muted (during sampling), no playback is audible and the sound at the audio input is switched into the output so that it can be monitored while sampling. The sound is picked up *after* the Input Sampling Filter so that the effect of the filter on the frequency response can be monitored. For proper operation, the CSTB signal is gated with the MUTE line by NOR gate U40c to form PAN CSTB. When the output is muted, this line is always low, allowing the input filter signal to pass all of the time. When the output is not muted, this line follows CSTB and provides the deglitching control.

Output Filters

Although we have been talking about the previous circuitry as analog in nature, it is really only quasi-analog since we were dealing with short pulses of analog signals. At some point these short pulses need to be turned into a smooth audio signal which can be sent to a mixer or PA system and that's what the output filters are for. In our previous designs, the Curtis filters not only provided dynamic filter effects, but were also used to smooth the audio output. Since we don't have Curtis filters anymore, output smoothing filters are required, but don't panic! These filters are much simpler and more reliable than Curtis chips. There are only two filters (one for each stereo channel), their frequencies are fixed and they operate at high signal levels, so there is none of the noise, distortion or popping we've all grown to know and love.

The filters are made up of stages. Taking the Right Output Filter, U60a is the first stage providing gain and one pole of low-pass filtering to the Deglitcher output. This is followed by a three-pole low-pass filter formed by U61b and a notch filter formed by U61a. The goal is to filter out as much of the output sample rate (reconstruction) clock as possible with the low-pass filter, then "notch" out whatever remains. This gets a little

complicated since the output sample rate changes depending on whether 12, 16 or 20 voices are active. The filters are carefully designed to provide the best compromise given the design criteria.

The Left Output Filter is identical to the Right using U60b, U61c and U61d. The output jack configuration allows a mono mix output, a stereo headphone output and separate left and right outputs for stereo operation, using only two jacks. With only the *Right/Mono* Output plugged in, left and right audio are mixed in mono at the Right Output via R63 and R81. With only Headphones plugged into the *Left/Phones* Output, the phones are driven by the separate left and right outputs via R130 and R142. With both jacks used, the switch in the Left jack breaks the mono mix connection, while the switch in the Right jack prevents the ring of the Left jack, which is connected to ground when a mono plug is used, from shorting out the right output. This provides separate left and right outputs for stereo operation. Diodes CR17 and CR18 protect U61d from damage by externally applied signals, as the 56Ω output resistor (R130) is too low a value to prevent this.

INPUT CIRCUITRY

The input circuitry consists of the *Input Preamp, Input Filter, Sample/Hold, Input MUX* and *A/D Converter* blocks as shown on the schematic.

Input Preamp

The sampling input jack (*Audio In*) feeds the input preamp (U49b), which provides a high input impedance, input overload protection (via CR6 and CR8), AC coupling to eliminate any DC offset and selectable gain for Line or Mic level. Gain selection is accomplished by the *MIC/LINE* control from the digital section. This line controls the remaining switch in U54 (the same 4053 used for the Deglitcher). The Preamp is set up for high gain at mic level by R89. When line level is selected, R111 is switched into the feedback loop by U54, reducing the gain for line level signals. The Preamp feeds the Input Sampling Filter.

Input Filter

All samplers require an Input Sampling Filter to prevent aliasing. Sampling theory requires that any frequency components greater than 1/2 the input sampling rate be removed from the input signal or aliasing will occur. The Input Filter on the EPS is a 36 dB/Octave chebyshev filter using an XR-1008 filter chip (U42). This chip is a self-contained *switched capacitor filter* whose cutoff frequency is controlled by an external clock. The clock (*FCLK*) is generated by the digital section. FCLK is programmable, allowing the cutoff frequency to be set appropriately for each input sample rate provided by the EPS. The only additional circuits required are an input filter and an output filter. What's that you say, the filter needs a filter? Well, yes. Switched capacitor filters and related chips such as analog delay lines are essentially *analog* sampling devices and require an input filter to prevent aliasing and an output filter to smooth the output.

The input to the XR-1008 is protected from overload by CR7 and CR9 and filtered by the two-pole passive network connected to pin 8. The output of the XR-1008 is smoothed by three poles of low-pass filtering consisting of a passive network and U49c. The output of U49c is DC shifted completely above zero by the voltage divider formed by R66 and R69 in order to drive the Sample/Hold. This output is also AC coupled into the Deglitcher (via C118, R115 and R116) so that the output of the filter can be monitored during sampling. R114, R155 and R156 are used to bias the Deglitcher switch inputs to prevent popping when muting and unmuting the output. The Input Filter feeds the Sample/Hold.

Sample/Hold

The purpose of the Input Sample and Hold circuit is to capture the voltage of the filtered input signal at the beginning of the sample period and hold it constant while it is being digitized by the A/D converter. The digitizing process takes 23 µS and the voltage being digitized must be constant during this period or errors will result as the A/D tries to track a moving target.

The S/H on the EPS is built in two stages. The first stage is a "sloppy" S/H that grabs the voltage at the beginning of a sample cycle and low pass filters it to remove any noise generated when the S/H switches. This first stage consists of a 4016 analog switch (U59) which closes to allow the signal to charge C109 through R120, then opens. C109 is buffered by U57a whose high input impedance prevents the capacitor from being discharged. While the switch is open, the output of U57a remains at whatever voltage C109 was charged to.

The control signal for the switch, *CA3S*, is buffered by open-collector inverter U51b. In order to turn the 4016 fully off, the control signal must be as low as it's negative supply. U51 cannot pull any lower than ground, therefore the negative supply of the 4016 must be at ground. In order to achieve maximum dynamic range through the 4016, it's positive supply is connected to +12 volts. Once again, in order to turn the 4016 fully on, the control signal must be as high as it's positive supply. The output of U51b is pulled up to +12 volts by R127 to insure this. The 4016 also cannot pass any signal lower than it's negative supply (zero volts), necessitating the offset added to the Input Sampling Filter output. CR12 protects the 4016 from any negative voltages at it's input.

The second stage of the Sample/Hold is almost identical to the first, using CR14, U56, C108, U57b, U51e and R108. It is driven by a control signal which occurs in the middle of the first stage control signal. In effect, the second stage samples and holds the first stage. The low pass filter (provided by R120 in conjunction with C109) used in the first stage is eliminated to allow faster aquisition of the signal by the second stage. Together, the two stages allow a much higher degree of isolation between the changing input and the held output than is normally possible with reasonable circuitry. The output of the S/H is attenuated and AC coupled into the Input MUX.

Input MUX

The Input MUX (Multiplexer) consists of a 4051 analog MUX chip (U53). This chip selects which of the analog inputs is to be digitized by the A/D converter. During sampling, the only input selected is the output of the Sample/Hold. During playback, the processor uses the Analog Select lines (*AN0, AN1* and *AN2*) to periodically select each of the various controllers (wheels, etc.). The voltage each controller puts out changes very slowly as compared with the audio input, therefore no Sample/Hold is necessary on these voltages. A person simply can't move the controllers fast enough to cause a significant change in voltage during digitizing.

Each analog input is scaled according to it's range. The Sample/Hold needs no scaling, but CR10 and CR11 protect the input from overload. The controller inputs are all level shifted above ground to minimize noise when they are being scanned. The Pitch and Mod Wheels (which run from positive to negative) are scaled by R106 and R107, then level shifted by R163 and R164. The patch buttons (which are positive going only) are scaled by R100. The Volume and Data Entry sliders (also positive only) are scaled by R104 and R105. The CV/Pedal input (which is negative going only) is scaled by R103 and level shifted by R101. R59 provides the supply for the pot in the CV pedal. An externally applied control voltage must range from -9 to 0 volts for proper operation.

The output of the MUX is buffered by U65a. The DOC reference voltage (+VREF) is also subtracted from the MUX voltage by this op-amp. This level shifts the final output to cover the range required by the A/D converter (which digitizes negative voltages only). There aren't enough pins on the DOC chip to bring out a feedback resistor for the A/D converter, so there is no way of guaranteeing the gain of the A/D. The gain could be externally trimmed to match the input signal, however we chose to trim the input signal to match the A/D, as this allows us to adjust both the gain and the DC offset with a single trimmer.

U65a provides sufficient gain to the MUX output signal to insure that the input to the A/D is too large. This allows us to increase or decrease the gain of the signal using Pot 3 (*A/D OFFSET*). This pot simultaneouly adjusts the signal gain and DC offset. Since the DC offset is a function of +VREF and this is the same reference signal used internally by the A/D converter, when the offset is set properly, the gain is automatically correct (and vice versa). What happens is, the circuit is adjusted by trimming the gain while checking for zero offset. When zero offset is achieved, the gain is right. Everybody's happy!

A/D Converter

The majority of the A/D converter circuitry is built into the DOC chip, however a number of external circuits are necessary for proper operation. The A/D converter is a conventional *Successive Approximation* design. A Successive Approximation Register (SAR) drives a 13-bit D/A converter, both of which are built into the DOC chip. An external op-amp (U62) is needed to convert the D/A output into a voltage and an LF311 comparator (U46) is needed to compare the D/A output with the input signal being digitized. The SAR sets the bits of the D/A to produce a voltage and this voltage is compared with the input. If the input is higher than the D/A voltage, the SAR increases the D/A voltage. If the input is lower than the D/A voltage, the SAR decreases the D/A voltage. The SAR homes in on the input voltage one bit at a time.

Actually, U62 sums the inverted output of the internal D/A with the MUX output. When these signals cancel out (i.e. produce zero volts at the output of U62), then they match and the conversion is complete. The output of U62 drives the comparator whose trip threshold is set up to be very sensitive around zero (actually about 1.7 mV) by R152 and R153. Diodes CR4 and CR5 limit the comparator input difference as the input can be several volts until the SAR really homes in. Voltages outside the threshold range aren't critical and would simply overdrive the LF311 input without the diodes. R154 provides hysteresis feedback to prevent the comparator output from "chattering". When the comparator switches, the output is fed back to the input to shift the comparator threshold even farther from the comparator input, locking the output state.

Basically what happens is, if the D/A output is greater than the MUX output, the output of U62 will be below the comparator threshold, keeping the comparator output high. This informs the SAR that the D/A voltage is too high and should be reduced. If the D/A output is less than the MUX output, the output of U62 will be above the comparator threshold, forcing the comparator output low. This informs the SAR that the D/A voltage is too low and should be increased. This comparison occurs for each bit of the D/A. As the SAR homes in, the output of U62 becomes smaller and smaller, which is why such fine sensitivity around zero is required.

The output of the comparator is latched by flip-flop U47b to form the *A/D CMP* signal which drives the SAR compare input on the DOC chip. This insures that the input will remain constant even if the small amount of noise generated by the input switching causes the output voltage of U62 to re-cross the comparator threshold. This would cause the output of the comparator to switch back, giving improper results.

As with the output D/A converters, the converter in the A/D requires an MSB trim to compensate for tolerance errors. Unfortunately, there are no pins left on the DOC chip, so the trim must be accomplished externally. Flip-flop U63 (both stages) is used to delay the output of the comparator latch so that the outputs of U63b switch at the time the MSB comparison is occurring. Pot 2 (*INPUT MSB ADJUST*) allows a small, variable amount of voltage to be summed into the output of U62 to compensate for MSB errors.

MISCELLANEOUS

Analog Timing

The remaining analog timing circuits consist of U40b which supplies an inverted CSTB signal for other circuitry and flip-flop U47a which, together with three open-collector inverters (U51c,d and f) create a pulse stretcher on CA3. CA3 is delayed by the flip-flop, whose output is wire-ored with CA3 by U51c and U51d to form CA3S. This signal falls when CA3 falls, but doesn't rise until one CSTB delay after CA3. This provides proper timing for the S/H and Input MSB circuitry. An inverted version of CA3 is also available from U51f to drive the Output Expander circuitry.

Output Expander Support

The Output Expander operates in the same fashion as the Panner and Deglitcher. Instead of generating 8 different pan positions, it produces 8 solo outputs. To accomplish this, the Output Expander requires the multiplexed audio output of the DOC, CA0, CA1 and CA2 to select the appropriate output channel, CA3 to determine whether the audio is meant for the Panner or the Solo outputs and CSTB to deglitch the output.

The multiplexed audio out is supplied in differential form (+SIGO and -SIGO) which, when combined by a Diff. Amp in the Expander, eliminates any noise picked up by the expander cable. CA1 is buffered and inverted by U51a and CA2 is buffered and inverted by U40d. CA0 and CA3 are handled differently as there are not enough pins on the Expander connector to bring out both signals seperately. Instead, both bits are combined into a single analog signal by U50c. This is a simple 2-bit D/A converter which encodes the four states into different voltages. These voltages are then decoded in the Output Expander to regenerate CA0 and CA3.

The deglitcher signal for the solo outputs (EX CSTB) is generated by U40a. This is an inverted version of CSTB during playback and is forced low by the MUTE line to mute the solo outputs during sampling (the input signal can only be monitored at the stereo outputs). The Expander also draws power (+12V, -12V and AGND) from the connector.

The EPS Tower Board

GENERAL

Due to variations in semiconductor processing of the DOCII chip from wafer lot to wafer lot, some DOC chips showed an unacceptable level of noise in the analog circuitry built into the chip. In order to eliminate this noise, additional circuitry was added. Since the problem didn't surface until after the main board PC layout was finished, it was necessary to incorporate this additional circuitry into a secondary board which mounts onto the main board. This *tower* board taps off points on the main board and replaces certain main board components with components on the tower board. Specifically, the following components are not populated on the main board:

R68	5.1	KΩ
R69	3	KΩ
R76	4.87	KΩ 1%
R77	4.87	KΩ 1%
C66	150	pF
C71	10	pF
C72	0.1	µF
C74	10	pF
C75	10	pF

The tower board circuitry consists of two parts, an output section and an input section. The output section consists of a dual VCA (voltage-controlled amplifier) which modulates the volume of the Waveform D/A converters with the DOC Volume output. The input section consists of an input compressor which amplifies low level signals to improve their signal-to-noise ratio. The sampling software automatically expands the sound data after sampling to restore the original dynamic range. This is essentially the same as companding.

OUTPUT CIRCUITRY

In the original design, the DOC Volume D/A output was converted into a voltage which fed the reference input of the Waveform D/A. This created a floating point output structure which allowed the volume of the waveform to be modulated by envelopes, etc. Unfortunately it was found on some DOC chips that as the Waveform D/A reference decreased, background noise became audible producing a "swishing" noise as sounds decayed. The solution was to always run the Waveform D/A converter at full reference, then follow the Waveform converter with a VCA to modulate the volume. The gain of the VCA is controlled by the DOC Volume D/A output. This produces the same result as the original design, but is quieter.

Due to internal connections existing in the DOC chip, there are some complications to the circuitry. The Waveform D/A reference is internally connected to the Volume Feedback input of the DOC chip (pin 16). Connecting this pin to -12V gives a full level reference to the Waveform D/A, but also adds a constant DC offset to the output of the Volume D/A which must be compensated for. Additionally, the Volume D/A output produces a voltage which decreases with increasing volume. This was fine for use as the Waveform D/A reference, but the VCA (MTA 1537) requires an increasing voltage to increase volume. On the tower board, U2 (pins 12, 13 and 14) along with feedback resistor R6 converts the current output of the Volume D/A into a voltage. U3 (pins 1, 2 and 3) then inverts this voltage for use with the VCA. R24 adds an offset to compensate for the DC offset in the Volume D/A output. Maximum VCA gain occurs at zero volts, so POT2 (GAIN) with R22 injects a variable DC offset into U3 to trim the maximum output to zero volts. R17 and R18 attenuate the final volume voltage to the proper level for driving the VCA. C4 replaces C71 on the main board.

The Waveform D/A outputs, when fed with a -12V reference, would produce positive only waveforms. The VCA requires a specific input bias network for proper operation, and this network assumes a bipolar input. The resistors in resistor pack RP1 add in an offset voltage and adjust the gain of the Waveform converter op-amps (U64b and U65b) to produce the desired bipolar outputs. C5 and C6 on the tower replace C74 and C75 on the main board.

The input network for the VCA is formed by RP4. R11 and R12 set the input bias voltage as required by the VCA. The current outputs of the VCA are converter to voltages by U3 (pins 12, 13, 14 and pins 8, 9, 10). Once again, the outputs are offset and gain-adjusted to use the full range of the op-amps. RP2 sets the offset and RP3 sets the gain. In addition, POT 1 (OFFSET ADJUST) with R19 allows any differential DC offset in the VCA outputs to be nulled out. The outputs of U3 are sent back to the main board where they feed the original Diff. Amp (U64a) and the Output Expander connector (J16). R9 and R10 on the tower replace R76 and R77 on the main board.

INPUT CIRCUITRY

The input signal chain is broken between the Input Sampling Filter output and the first Sample/Hold. A Compressor circuit is added on the tower (U1, pins 5, 6 and 7). The compressed input signal is mixed with the original signal to minimize any gain tracking errors in the compressor, then the signal is fed back to the first Sample/Hold on the main board.

R2 and C1 on the tower simply replace those components on the main board which had to be removed to gain access to the necessary signal points (R68 and C66). On the main board, R69 is eliminated and C67 is replaced with a jumper in order to ground the input of U49c (pin 10). This eliminates the DC offset which was added to the op-amp output for driving the Sample/Hold. This offset is added in later on the tower board.

The compressor is a common design using an op-amp (U1, pins 5, 6 and 7) and based around the non-linear conduction characteristics of diodes. The circuit initially has high gain, so low level signals are amplified. As the input signal approaches the conduction threshold of diodes CR1 and CR2, they begin to conduct, reducing the gain of the circuit. The conduction curve of a diode is exponential, therefore signals are exponentially compressed. As signal level increases, gain decreases.

The compressed signal is mixed with the direct signal by U1 (pins 1, 2 and 3) to reduce errors caused by temperature drift in the compressor diodes. After sampling, software processes the compressed data samples through an expansion algorithm which reverses the effect of the diodes. That is, samples are scaled using a digital model of the diodes to accurately restore the original dynamic range of the signal. R15 is used to add an offset to the final output of the compression circuit so that it can directly feed the Sample/Hold circuit.

Because the conduction characteristics of semiconductor junctions are temperature sensitive, a "heater" circuit is built-in. The circuit, consisting of a half-watt resistor (R3) directly across the power supplies, is used to heat the diodes and keep them at a constant temperature independent of the ambient temperature.

These circuit modifications do not simply compensate for "noisy" DOC chips, they insure that the EPS achieves the full fidelity potential of its encoding scheme. The EPS can meet its' specifications, unlike many sampling and digital processing products which claim 12 or 16-bit resolution, yet measure well below their theoretical performance limits due to marginal analog circuit design.

New EPS Main Boards (10001, 10002)

The new EPS main boards incorporate a number of refinements to the original design. The circuitry in both the digital and analog sections has been improved and simplified to provide superior performance and increased reliability, while parts count and power consumption has been greatly reduced.

DIGITAL SECTION

The major change in the digital section involves the use of a custom Gate Array integrated circuit (U36) to replace most of the random logic in the original design. This 68-pin Quad-Inline-Pack (*QUIP*) replaces the three 82S153 PLAs (the *ICU*, *BCU* and *MMU*) and the 74LS138 Decoder (U32) used for I/O decoding. It also replaces the 74LS74 Flip-flop used to demultiplex DA18 (U30b) and delay the FDC DTACK response (U30a). The 74LS74 Flip-flop (U34) used to generate the 5 MHz and 500 KHz clocks is replaced, as are the 74LS161A Divider (U37) and 74LS74 Flip-flop (U39) used to generate the programmable clock for the Input Sampling Filter. All of the input and output signals in the new design correspond to signals of the same name in the original design and are functionally identical, therefore the original documentation concerning these signals is still applicable. Certain signals which are now completely internal to the Gate Array are no longer listed.

The Gate Array has many advantages. The 68-pin QUIP takes up less PC board space than the original design, resulting in a smaller and less expensive PC board. The Gate Array is fabricated with CMOS technology, greatly reducing power consumption which, together with the reduction in parts count, leads to greater reliability. Finally, the Gate Array is less expensive than the integrated circuits it replaces. Although the Gate Array is initially being socketed, when we have the capability, we will surface mount this package onto the main board.

The only difference between the 10001 and 10002 boards is the memory configuration. The 10001 board is layed out to take 256K x 1 DRAMs in a ZIP package, while the 10002 board is layed out for 256K x 4 DRAMs in a ZIP package.

ANALOG SECTION

There are a number of changes in the analog section. First, as a result of the board space gained by the use of the Gate Array in the Digital Section, all of the circuitry which was originally mounted on the Tower board has been incorporated onto the main board. This simplifies the manufacturing process and eliminates the Tower board completely.

The -5A supply and +8.75 volt Reference supply, originally created using op-amps U49a and U49d are now generated using discrete regulators Q5 and Q3 (Q3 has a diode in the ground leg to bring the output voltage up to 8.7 volts from the nominal 78L08 output of 8 volts).

Additionally, the Input Sample/Hold circuitry was greatly simplified using a more efficient design. The two stage Sample/Hold in the original design (U59, U57a, U56, U57b) has been replaced by a single stage design consisting of U57 and U50a. The circuit is configured to store the sampled voltage across C128. Because the analog switch (U57b) is incorporated into the feedback path of the op-amp (by R190), the circuit is able to track input voltages much faster. Since U57b switches between virtual ground and true ground in a current-mode configuration, any non-linearities in the switch are minimized, reducing distortion. Isolation between the input and output of the circuit is greater since the input signal is grounded when not being passed to the output. The circuit is also bipolar--signals swing plus and minus around zero--unlike the old design which had an offset to keep all signals positive. Since this offset is no longer needed, the offset resistor has been eliminated (R15 on the tower).

The Input DC Offset adjust has been separated from the Input Gain adjust. Originally, POT3 was used to adjust both the input gain and DC offset simultaneously. These adjustments didn't always track accurately after the components aged in burn-in. In the new design, POT5 provides gain adjust in the same fashion as the old circuit. DC offset is compensated for by POT2 which can inject a variable DC voltage into the signal being converted. Q4 is used to buffer the voltage generated by POT2 and provide a low impedance source. This allows the pot to be located away from the critical circuitry and out from under the keyboard so that it can be easily adjusted after burn-in.

The input MSB adjustment has been modified to provide more range by level shifting the output of the MSB Flip-flop. In the original design, the adjustment pot (POT2) was driven directly off of the MSB Flip-flop (U63b). In the new design, the 0 to 5 volt output swing of the MSB Flip-flop (U58b) is level-shifted to a range of -5 to +5 volts by U51b and U51c. The adjustment pot (POT6) is tied across these shifted signals.

In the output section, termination resistors (R204, R207 and R202) have been added to the inputs of the Panner (U56). This allows a hotter signal to be driven into the panner without overloading it, thereby improving the signal-to-noise ratio. This termination to ground also reduces switching noise. The same technique is used on the Deglitcher (R196 and R194 on U54).

The final change is in the Analog Timing section used to drive the OEX-8 Output Expander. The new Sample/Hold circuit eliminated the need for much of the analog timing circuitry, allowing the 7406 pack (U51) to be eliminated. Since some of the inverters from that pack were used in the output expander circuitry, a re-arrangement of the circuitry was necessary. CA1 is now buffered and inverted by Q6. CA3 is no longer present in an inverted form to drive the inverting input of the combiner circuit (R73 and U50c of the old circuit). Instead, the true version of CA3 is fed into the non-inverting input of U48b via R182. As in the previous design, U48b acts as a simple 2-bit D/A converter which produces different output voltages depending on the states of CA0 and CA3. The voltages are decoded by the OEX-8 Expander to re-create CA0 and CA3.

The EPS Display/Keypad Board

GENERAL

The Display board for the EPS is very similar to the ESQ-1/SQ-80 display board in operation, but is much simpler as a more efficient microcomputer is used. It uses the same Vacuum Fluorescent technology, based on a custom FIP (Fluorescent Indicator Panel) manufactured for us by NEC. The 7538 microcomputer (U3) is also manufactured by NEC and is used to update and control the display, scan the keypad for button presses and light the LEDs. The 68000 tells the 7538 what messages to display and what LEDs to light and the 7538 informs the 68000 what buttons have been pressed.

FIP DETAILS

The FIP is basically a large vacuum tube just like those in the old days. It has a filament which emits electrons, grids which attract electrons and anodes which control the flow of the electrons. What makes the tube a FIP is the fact that the anodes are coated with a fluorescent material which glows in a characteristic blue-green color when struck by electrons. In the case of the FIP, the grids are called *Digits* and the anodes are called *Segments*. This is because the grids are used to control which digit of the display is lit at any given point in time and the anodes are used to control which segments are lit for that digit (thereby determining what alphanumeric character is shown at that digit position).

The FIP has 22 alphanumeric character positions (digits) and two "indicator" areas. Each character position has 16 segments (including the underline cursor and the decimal point). This allows the display to generate all the characters of the alphabet, the numerals 0 through 9 and a number of punctuation symbols. The indicator areas are made up of tiny words which can be lit individually to indicate, for example, what mode the EPS is in. These words are similar to the "day of the week" indicators on a VCR. The indicators may *look* completely different from the alphanumeric characters, but they are treated exactly the same way. Actually, each of the two indicator areas is a digit and each of the little words is a segment of that digit.

FIP OPERATION

In order for a particular character to appear at a particular location, that digit must be "on" (at a high voltage) and the appropriate segments that make up the desired character must be "on" (at a high voltage). The ESQ-1 display used 55 VDC as high voltage, while the more efficient EPS FIP requires only 40 VDC. Unlike the ESQ-1 display which used all positive voltage, the EPS uses both positive and negative voltage to drive the display. To turn a digit or segment off, the appropriate line is brought to -35 VDC. To turn a digit or segment on, the appropriate line is brought to +5 VDC. The total voltage swing is -35 to +5, or 40 volts. Both of these voltages are generated by the power supply.

The electrons which ultimately cause the display to light are produced by the filament, which is driven by 5 VAC, also generated by the power supply. To cause enough electron flow to light a segment, the digit and segment voltages must be higher than the filament voltage (the bigger the difference in voltage the brighter the display). In order for a digit or segment to be shut off, the digit or segment voltage must be lower than the filament voltage. To accomplish this, the filament is biased to a DC level of 6.5 volts above the "off" voltage (-35 VDC). That is, the filament voltage generated by the power supply is a 5 VAC signal biased to a DC level of -28.5 VDC. The turn off voltage (-35 VDC) is sufficiently lower than the filament (by 6.5 volts) to ensure shut off, while the turn on voltage (+5 VDC) is sufficiently higher than the filament (by 33.5 volts) to insure a bright display.

The display itself is scanned. That is, each digit is only on for a short period of time. Since only one digit is ever on at any given point in time, the segment lines can be shared between all digits, therefore there are only 16 segment lines total, along with 24 digit lines (22 characters and 2 indicator areas). To scan the display, the segment information for the first digit is placed on the segment lines and the first digit is turned on. After a fixed amount of time (about 420 µS) the first digit is turned off, then the segment information for the second digit is placed on the segment lines. Next, the second digit is turned on for the same amount of time as the first. The process is continually repeated until all digits have been scanned, then the scan starts over again with the first digit. This process happens so fast that the human eye perceives the display as being constantly on.

LED OPERATION

The LEDs on the display board are treated exactly like the FIP, that is, they are scanned as a digit, with the segment information being used to determine which LEDs are lit. The LEDs are driven by a pair of 74LS374 latches (U1, U2) which latch the segment information at the appropriate digit time and hold it constant for the entire scan cycle. This is necessary as LEDs aren't as efficient as FIPs and are too dim when lit for only 420 µS. The resistors in series with the LEDs set the operational current (and therefore the brightness) of the LEDs. The red LEDs are set to less current than the yellow LEDs as yellow LEDs are dimmer than red ones. The yellow LEDs are lit as brightly as possible and the red LEDs are adjusted to match the yellow.

The latches are driven from the same segment lines which drive the FIP. These lines swing from -35 to +5 volts, which would damage the TTL inputs of the latches if connected directly. In order to insure proper operation, each latch input is protected by a diode. The diode allows only the positive (+5 volt) signal level through, which is the normal TTL high level. As soon as the control line goes negative, the diode reverse biases, preventing any negative voltage from reaching the TTL input. When this happens, the 2 KΩ pull down resistor pulls the input to ground, the normal TTL low level. The digit drive line which strobes the segment data into the latches is protected by the same means, however a 1.5 KΩ pull-down resistor is used as this line must drive two inputs (one per latch) and must therefore sink more current.

KEYPAD OPERATION

The keypad buttons are handled in the same manner as all of our products, that is they are arranged in a matrix of rows and columns with each button occupying a unique position in the matrix. On the EPS, there are 6 columns and 6 rows for a total of 36 buttons. The columns are outputs of the 7538, while the rows are inputs. The 7538 scans the keypad by bringing one of the column lines low, then examining each row input. If the 7538 sees a low row input, then a button is being pressed and it determines which button by what column and row intersection it represents. This process is repeated for each column and row, then it starts over again. The column outputs of the 7538 have no pull-down devices, so 3.3 KΩ pull-down resistors are used to set the proper low level. The 33 KΩ pull-up resistors on the row inputs insure that a high level is read when no buttons are pressed.

7538 MICROCOMPUTER

The 7538 controls the entire EPS display just as the 6500/11 controls the ESQ-1 display. The 7538 has a number of features which make the system simpler, including a built-in RESET circuit, built-in zero-crossing detector for line frequency synchronization and the ability to directly drive the FIP without requiring external high-voltage buffers. The 7538 communicates with the 68000 via a synchronous serial port. This means that there is not only a data in and a data out line, but also a clock signal which tells when the data is valid.

Timing

The master clock (CLK) for the 7538 is generated on the EPS main board (it is actually the CSTB signal from the DOC II chip). This 625 KHz clock is cleaned up by the 74LS14 schmitt triggers (U5) which also provide both clock phases as required by the 7538. The 7538 is a CMOS microprocessor and requires higher drive levels than the 74LS14 produces, so 3.3 KΩ pull-ups are used to insure proper levels

Reset

The RESET circuit is built into the 7538 and only an external resistor and capacitor are needed. The values used provide a RESET time constant of about 6 mS.

Line Synchronization

The scanning of the FIP is synchronized to the frequency of the AC line voltage. This is done for a number of reasons. The filament voltage is simply a stepped down version of the line input and therefore operates at the line frequency. If the scanning were not synchronized to the line, a beat frequency (the difference between the line frequency and the scan frequency) would be generated which would cause the display to flicker. Since all electric lighting operates at line frequency, this also eliminates flicker when the display is viewed under lights. Deriving the synchronization signal from the line input has the advantage of eliminating flicker regardless of the actual line frequency. That means we can use the exact same display circuitry in Europe (which uses 50 Hz line frequency) as we use in the US (which uses 60 Hz line frequency) and we don't have to worry about changing jumpers or component values.

The line sync signal is tapped off of the FIP filament. It is AC-coupled to remove the -28.5 VDC bias, clipped by diodes to protect the 7538 input from overvoltage and low-pass filtered to remove any noise or spikes that might be present. This signal is fed to the 7538 zero-crossing detector (INT0) which is used to generate an interrupt to the processor at the line frequency. When the 7538 receives a line interrupt it begins its scan. It scans all digits and, when finished, blanks the display and waits for another line interrupt to tell it to start again. In the US, the display is scanned at a 60 Hz rate, which means the entire scan takes only 16.7 mS.

High Voltage Drivers

The output pins of the 7538 are capable of directly generating the FIP drive signals which swing from -35 to +5 VDC, without the use of the external buffers needed on the ESQ-1. For proper operation, the 7538 must have a +5 volt supply, a -35 volt supply and a bias supply. This bias supply is generated using a 9.1 volt zener diode between the -35 and +5 volts. This produces a bias voltage which is 9.1 volts (+/- 5%) below the 5 volt supply (approx. -4 volts).

Synchronous Communications Port

Like the ESQ-1 display, the EPS display communicates with the main microprocessor over a serial port. The port on the EPS is a synchronous 8-bit interface. That means it transmits 8 bits of data out (on the MOSI line) while simultaneously receiving 8 bits of data in (on the MISO line) synchronized to a clock signal (on the SCK line). These lines are buffered by the remaining schmitt triggers (the data in and out is inverted, which is easily compensated for in software, but the clock signal must retain proper phase relationship with the data to insure accurate data transfer).

These lines interface with a similar synchronous port built into the 68HC11 microcomputer on the poly-pressure KPC board. Since the 7538 generates the clock, it is designated the master and the 68HC11 is the slave. Whenever the 68000 wishes to display a message, the message is sent via the DUART on the EPS main board to the 68HC11 on the KPC board. The 68HC11 then sends the message along to the display via the synchronous port. Likewise, when a button is pressed, the 7538 sends this information to the 68HC11 via the synchronous port. The 68HC11 then sends the information to the 68000 via the DUART. This is necessary as the EPS main board has no synchronous port while the KPC board does.

58341 HIGH VOLTAGE SHIFT REGISTER

The 7538 does not have enough outputs to drive all of the segments and the digits as well. In order to generate the digit drive signals, a 58341 shift register (U4) is used. This device acts like any other shift register except it has built-in high voltage output divers which are capable of swinging from -35 to +5 volts. Since only one digit is ever on at a time, a shift register is ideal as it allows a single one (on) bit to be shifted through easily.

The 7538 controls the shifting of the on bit using the Data In line (DIN), the Clock line (CLK) and the Enable line (EN). The 7538 uses the Data In line to load a single one bit into the shift register at the beginning of a scan. The Clock line is used to shift the on bit to the appropriate digit position. The Enable line is used to enable the digit output once the on bit is in position. The Enable line is also connected to the Blanking input (BLANK) of the 58341 so that the display is blanked (off) whenever a new digit is first set up. This prevents smearing from character to character when the segment information changes. The blanking input is also used at the end of a scan to turn off the display until the next line frequency interrupt occurs to restart the scan. If the display were left on at the end of a scan, the last digit would appear brighter than the rest as it would be on fractionally longer than any other digit. These lines have no active pull-downs, so once again 3.3 KΩ pull-down resistors are used to insure proper levels.

The two indicator areas of the FIP are much larger than the other digits and therefore require much more current to light them. The 58341 can't supply enough current for these two digits, so the outputs are buffered by emitter followers Q1 and Q2. These transistors simply provide more current drive.

SLIDERS

The Volume and Data Entry sliders provide variable voltages from zero to five volts depending on the slider position. These voltages are sent to the analog section of the EPS main board were they are digitized by the DOC II.

An Overview of the EPS Power Supply and Patch Select Buttons

Power Supply

The power supply is similar to our previous designs, in fact, it uses part of the ESQ-M design together with circuits common to all of our supplies. The Analog supply is simple. The center-tapped transformer secondary (fed through protective fuses F1 and F2) is full-wave rectified by CR3, CR4, CR5 and CR6 to produce positive and negative unregulated supplies. These supplies are filtered by C8, C9, C16 and C20, producing approximately +/- 18 VDC which are fed over to the Main board as +VA and -VA. The supplies are then regulated on the Main board. +VA is also fed to Q4 to produce regulated +12 VDC for the Disk Drive. C17 and C18 filter this output.

The digital supplies are derived from the center-tapped transformer secondary (fused by F3 and F4) rectified by CR1 and CR2. This unregulated supply is filtered by C7, C13, C1, C4 and C10 producing from +7 to +11 VDC which is fed over to the Main board as +VD (where it powers the KPC board regulator). +VD also feeds the three +5 VDC regulators (Q1, Q2 and Q3) which generate +5D (used by the Main board digital section and the optional SCSI port), +5X (used by the Disk Drive and Display board) and +5M (used by the Main board memory section and optional memory expander). Each regulator has two output capacitors for filtration. The current consumption of the EPS with all expanders installed is higher than our previous products, necessitating an additional heat sink and the use of 4 Amp fuses (rather than our standard 2.5 Amp fuses) for F3 and F4. The +5D output is also brought over to the Main board as +5A. This separate line provides a "clean" +5 VDC to the analog section as little of the digital noise that +5D picks up on the Main board is transmitted back to the Power Supply board.

As with the ESQ-1, SQ-80 and ESQ-M, the filament supply for the FIP display is provided by a separate transformer winding fused by F5. The FIP used in the EPS requires 5.5 VAC, which is generated directly by the EPS transformer without need for the dropping resistors used on the ESQ-M (the filament winding on the ESQ-M transformer was designed for use with the ESQ-1 FIP, which required 9 VAC rather than 5.5 VAC). The filament bias voltage is generated in exactly the same fashion as on the ESQ-M. The -35 VDC supply is dropped through a 6.0 volt Zener diode (CR10) loaded by R5 and filtered by C19. This voltage is fed to the center-tap of the filament winding to level shift the filament AC voltage to 6 VDC above the negative supply.

The -35 VDC supply is generated almost identically as on the ESQ-M. One of the 26 VAC windings for the Analog supply is fed to a doubler circuit made up of C14, CR9, CR7 and C15. C14 and CR9 level shift this AC sine wave so that it is completely negative going, then CR7 rectifies the signal and C15 filters it to produce around -30 VDC (unregulated). R1 is a bleeder resistor to discharge the supply when the EPS is turned off. This supply doesn't really need to be regulated for the Display since any ripple in the supply will be at the line frequency. As the Display is sync'ed to the line frequency, the ripple will never cause any flicker.

The output circuitry on the -35V supply provides current limiting and overvoltage protection. The current limiting prevents the supply or display from being damaged should the display fail. The current is limited to an amount incapable of burning out the display or overloading the supply. As output current consumption increases in a failure mode, the voltage drop across R4 increases. This causes Q6 to turn on, pulling current from the base of Q5 through R3. This reduces the base voltage of Q5 and limits the current Q5 can supply. Normally, the voltage drop across R4 is small and Q6 is off. The overvoltage protection consists of Zener regulator R2 and CR8. If the voltage at CR8 ever exceeds it's Zener voltage (-36 volts), CR8 will clamp it to -36 volts. Note that in normal operation CR8 is <u>not</u> conducting. It only comes into play when abnormally high line voltages cause the negative supply to exceed the safe operating range of the Display board.

Patch Select Buttons

Oh come on, they're just two switches and two resistors! The resistors are different values which, together with the load resistor on the Main board (R100), form a two-bit D/A converter. Depending on the four switch states (neither pressed, switch 1 pressed, switch 2 pressed or both pressed) four different voltages are produced at the Input MUX.

Hints on Debugging the EPS Main Board

Note: This document is intended more for technicians who are debugging boards from the assembly line rather than customer service returns. Few of the problems described are likely to occur in the field.

NORMAL OPERATION

On power-up, you should see the RESET line go low for at least 900 mS. The SUPER line should go low after RESET and the OSROM chip select, URD, LRD, AS and DTACK should be moving. During boot-up, the processor is used to load the O.S. into the system, therefore the DMA control lines should not be low or moving and no interrupt lines to the ICU should be moving. You should, however, see the IRQ and DRQ lines on the FDC moving. You should also see the chip selects to the DUART and FDC moving as the processor accesses them, which implies that I/O and DS should also be moving. OSRAM will move as the data is loaded in.

The processor uses the DRAM on power up, so you should see SRAM, U245EN and L245EN move. R/W, UWE and LWE should also move during boot-up. DBUS, RAS, CAS and AMUX should be moving all the time and you should see CAS0 moving.

After normal boot-up, you should see the DUIRQ line moving at a frequency of around 1 KHz. Any other DUART interrupts (MIDI, keypad button presses, etc.) will also show up on this line.

When loading a sound, or performing any disk operation which uses DMA, you should not only see the disk lines moving, but the DMA control lines and DMIRQ should be moving as well. The SREQ line should always be low at these times. During a DMA transfer, the OWN line, DBEN and UAS should move.

When sampling, you should see the DOCIRQ line moving. You should also see the SREQ line moving at the sample rate, but the DRQ line from the FDC should be low. As above, the DMA control lines should also be moving.

DEBUGGING

Well, if everything was working right, you wouldn't be debugging the board so here are a few tips we learned when debugging the first 40 some boards. Probably the most important aspect of debugging is experience with the board. The more boards you work on, the more you'll learn how the signals in a properly operating system should look. You can often debug boards by hunting for signals that just don't "look" right.

The EPS has some built-in diagnostic routines that test the communications paths between the Main board, the KPC board and the Display board. If you turn the system on and don't get the start-up message, listen to the audio output. If you hear a beeping tone, it indicates that the Main board couldn't communicate properly. Assuming you are using a known good KPC, Display board and cables, the problem must be in the Main board. First check that the appropriate clocks are being applied to IP5, IP4, IP3 and IP2. Next, examine all of the DUART lines for anything that looks wrong -- lines stuck low, high or floating or lines that look shorted to other lines. We had one communications failure due to a data bus pin being bent under on the DUART, which prevented the DUART from being programmed properly.

Of course, you'll be lucky if you get the beep-tone at all on a bad board. By far, the most common failure mode we saw was open traces on the board. This resulted in the system immediately locking up, with the processor trying to access an illegal area of memory, such as $800000 - $BFFFFF. In this case, the AS line will always be low and the DTACK line will always be high as the processor waits for a DTACK that will never come (welcome to the Twilight Zone). Should you get a board which comes up in this state, look at all the lines to each DRAM chip for something that looks wrong. Nearly every board we had that came up in this state had an open trace to the DRAM section, either to the RAMs themselves, the address multiplexors or the data buffers. Most of these problems could be found with a scope and presumably between the GenRad and the Fluke, these problems will be a lot easier to find.

We debugged three batches of boards. The first batch of boards had not been tested for opens and shorts. They were awful. Maybe two out of ten boards came up and the rest took a long time to debug using the in-circuit emulator to find the shorts.

The second batch of boards were supposedly tested by the manufacturer for both opens and shorts. Although there were no shorts, there were plenty of opens. Maybe 4 out of 15 boards came up without problems, but the opens were relatively easy to find using a scope.

The third batch of boards was perfect. There were no opens or shorts and 13 out of 15 boards came up immediately with no problems. One board had a bent under pin on the DOC chip. The board came up with the start up message, would boot from the disk and would then display a crazy message with lots of asterisks. The problem was traced to the DOCIRQ line being constantly low. There was no problem with the interrupt line though. One of the DOC data bus pins was bent under and not contacting the data bus, which prevented the processor from programming the DOC properly. The other board had a chip in backwards which pulled down the +5 Digital supply, so nothing was working. Remember that there are two +5 volt supplies on the board -- one can be shorted while the other could be okay. The +5 Digital supply can be checked on pin 20 of any of the PLAs, while the +5 Memory supply can be checked on pin 20 of the 74LS245's.

Other problems we saw included bad or un-programmed PLAs in the board, one bad 68000 processor, wrong components in the RESET circuit and an open trace to the pull-up resistor on the BERR line which caused the processor to generate constant RESET pulses. Don't forget to check that unused lines with pull-up resistors are actually pulled up! There was also one problem with the FDC DREQ line to the DMAC. It caused the "DISK DMA IRQ ERROR" message to appear as the disk timed out without the DMAC ever receiving a data request. Remember that the processor, not the DMAC, is used to load the operating system, so it is possible for the system to boot up but still have problems loading sounds or overlays (which do use DMA). Pressing the "SAMPLE" button will load the sampling overlay, a convenient way to check if DMA is working.

Those are about the only tips we have at this point. As you develop your own set of hints, they should be added to this list. Good luck and happy debugging!

Hints on Debugging the EPS Display/Keypad Board

Note: This document is intended more for technicians who are debugging boards from the assembly line rather than customer service returns. Few of the problems described are likely to occur in the field.

The EPS display board is fairly easy to debug due to the small number of components used. Make sure all voltages are present. If both the +5 and -35 supplies aren't present, the CMOS chips can become reverse biased and act very strangely. Also verify that the bias supply on pin 4 of the 7538 is around -4 volts. If it isn't, something is wrong with the zener supply.

If the master clock is present, upon power up the LEDs should all light. Verify that a 625 KHz clock is present on both pins 2 and 3 of the 7538. These clocks should be square waves which go to 5 volts when high.

Verify that a 60 Hz waveform is present on pin 41 of the 7538. This waveform should appear as a smoothly rounded signal with the bottom clipped flat. It should range from around -.8 volts when low to around 3 volts when high. If this signal isn't present, the display will never scan.

If all of these signals are present and all of the LEDs don't light up, check the Reset input when you power up. There should be a positive going pulse which decays away to zero over about 9 milliseconds each time you turn on the power.

Once the display is scanning, you should see each keypad column line going high and low. The row lines should all be high unless you press a key on that row, at which point you should see the row line move along with the appropriate column line.

You should see a periodic burst of 8 low going pulses on the SCK line (pin 40 of the 7538, also check for it at pin 3 of J2, the Display Connector). MOSI (pin 39) should show a single positive going pulse during the same interval (also check for the inversion of this signal at pin 2 of J2). In a normally operating EPS system, you should see the same signal at the MISO input.

There should be a single positive pulse every 16.7 mS on the DIN line of the 58341 (pin 22). There should be 25 pulses every 16.7 mS on the CLK (pin 21), EN (pin 24) and BLANK (pin 25) lines. Each of the digit outputs of the 58341 should have a single pulse every 16.7 mS. The pulse should go from around -35 volts when low to around 0 volts when high. Depending on what characters are being displayed, you should also see the segment drive lines from the 7538 covering swinging from around -35 to about +4 volts. The segment lines will move much more often than the digit lines.

Verify that the inputs to the latches range from about .8 volts when low to around 3 volts when high. If a latch output is low, the corresponding LED should be lit.

Of the 50 or so displays we've built so far, we have had no problems other than a few open traces which prevented certain keys from working. If you have a board with keys that don't work, this is the most likely cause. Check both sides of the switch. One side should show a pulsing column line and the other side should be at 5 volts. When you close the switch, both sides should pulse. If a group of six switches are inoperative, there may be a problem with a particular row or column (shorts, opens, etc.). Shorts on the row or column lines can also produce bogus keypresses or cause improper keys to be sent when a button is pressed.